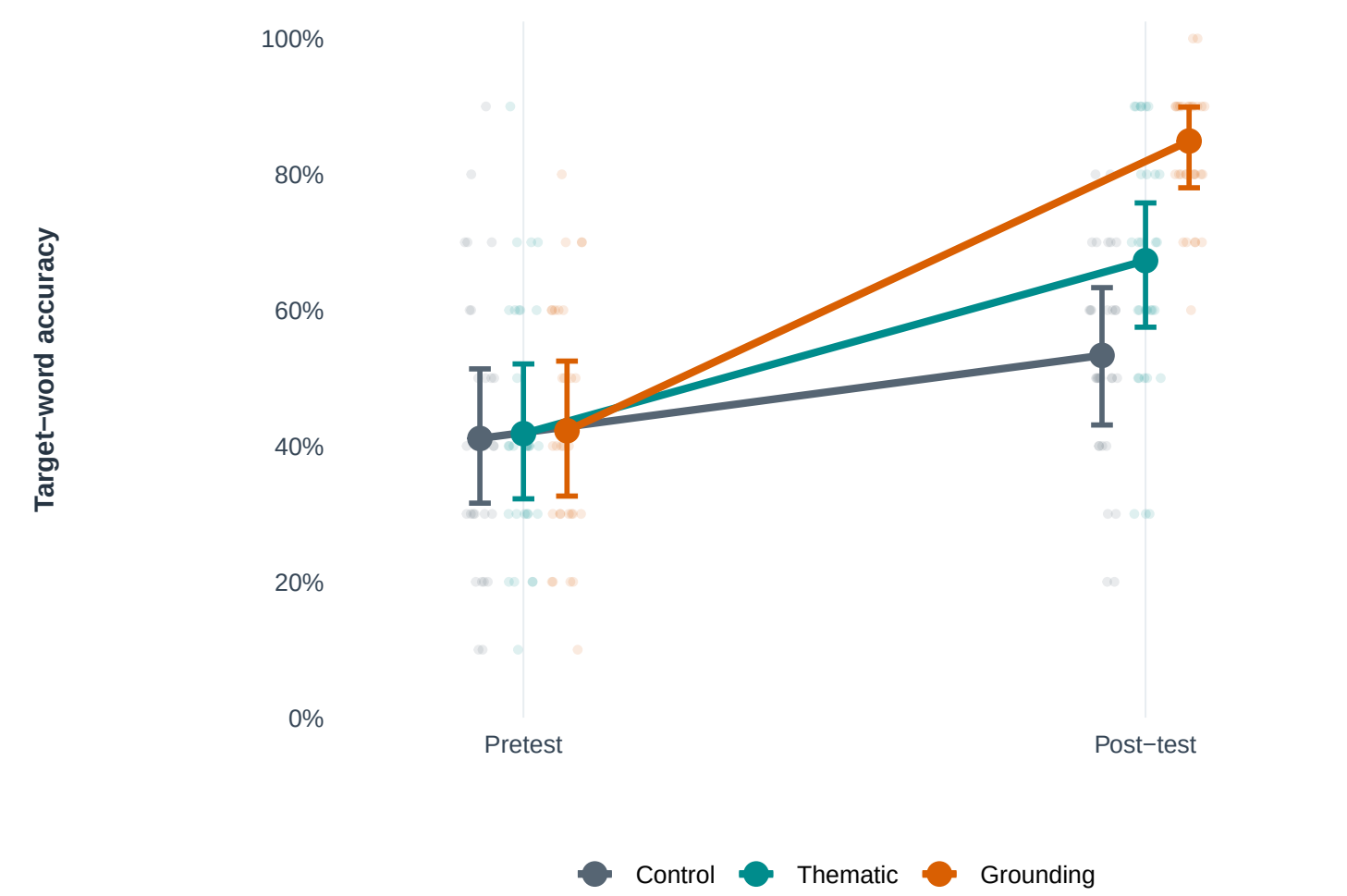


How vocabulary organization changed target-word learning

Primary results and robustness analyses

A Grounding produced the largest gain

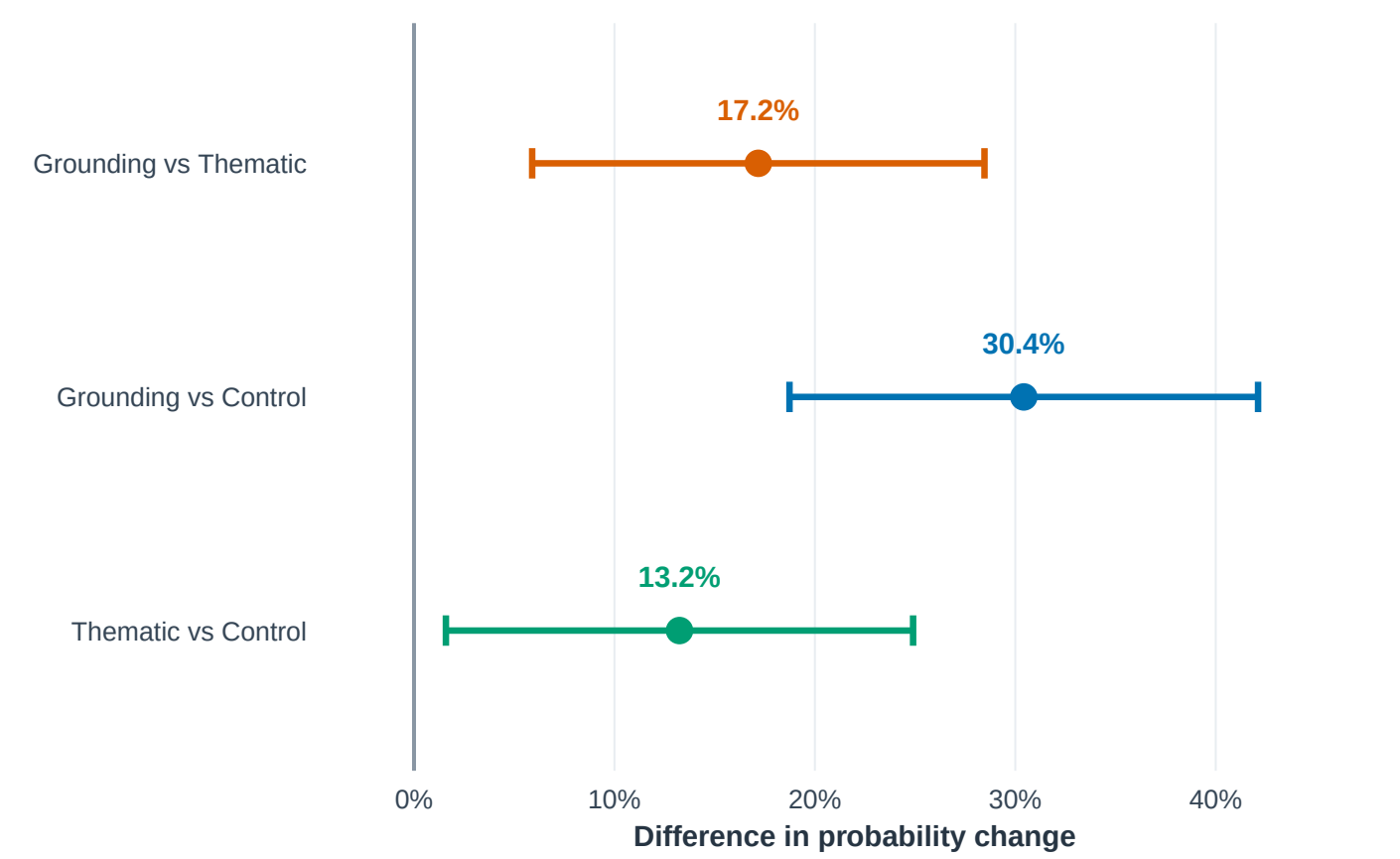
Model estimates and 95% CIs; faint points are participant scores



The overall time effect also contains the shared change in question format.

B Planned differences in learning gain

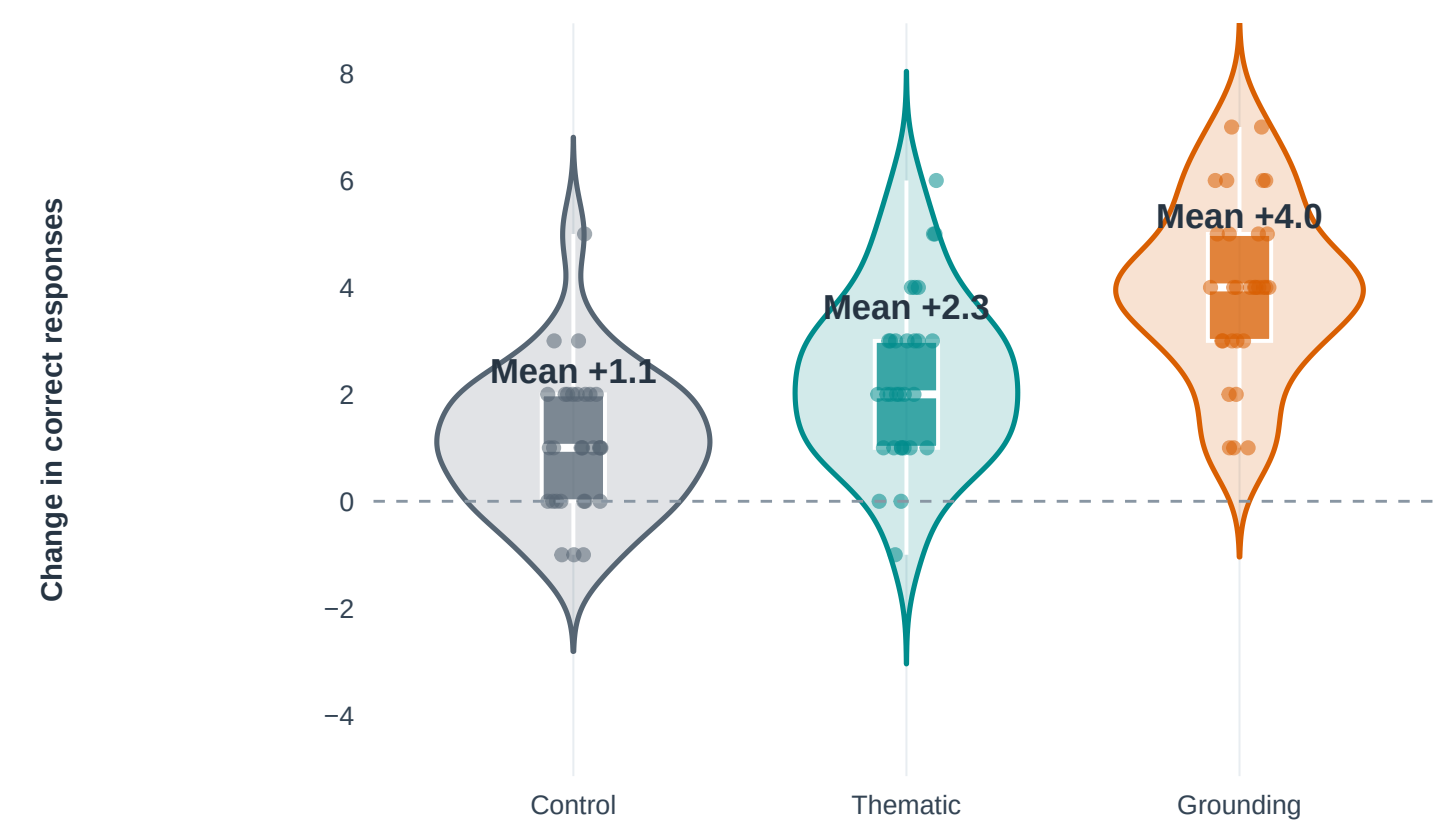
Difference in model-predicted pre-to-post change



Positive values favor the condition named first. Intervals are 95% confidence intervals.

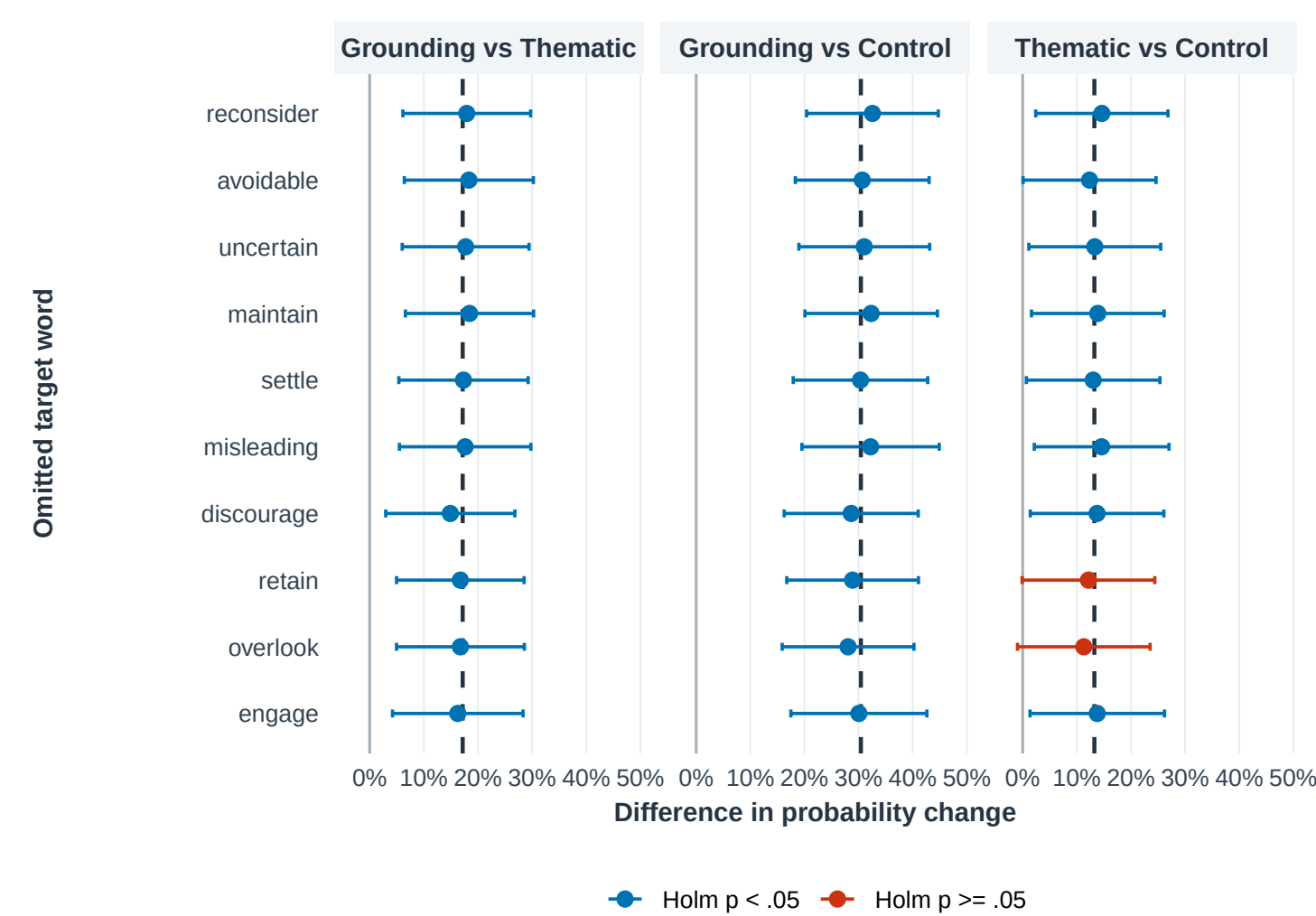
C Participant gain distributions

Post-test minus pretest correct responses on 10 shared target words



D Leave-one-target-out sensitivity

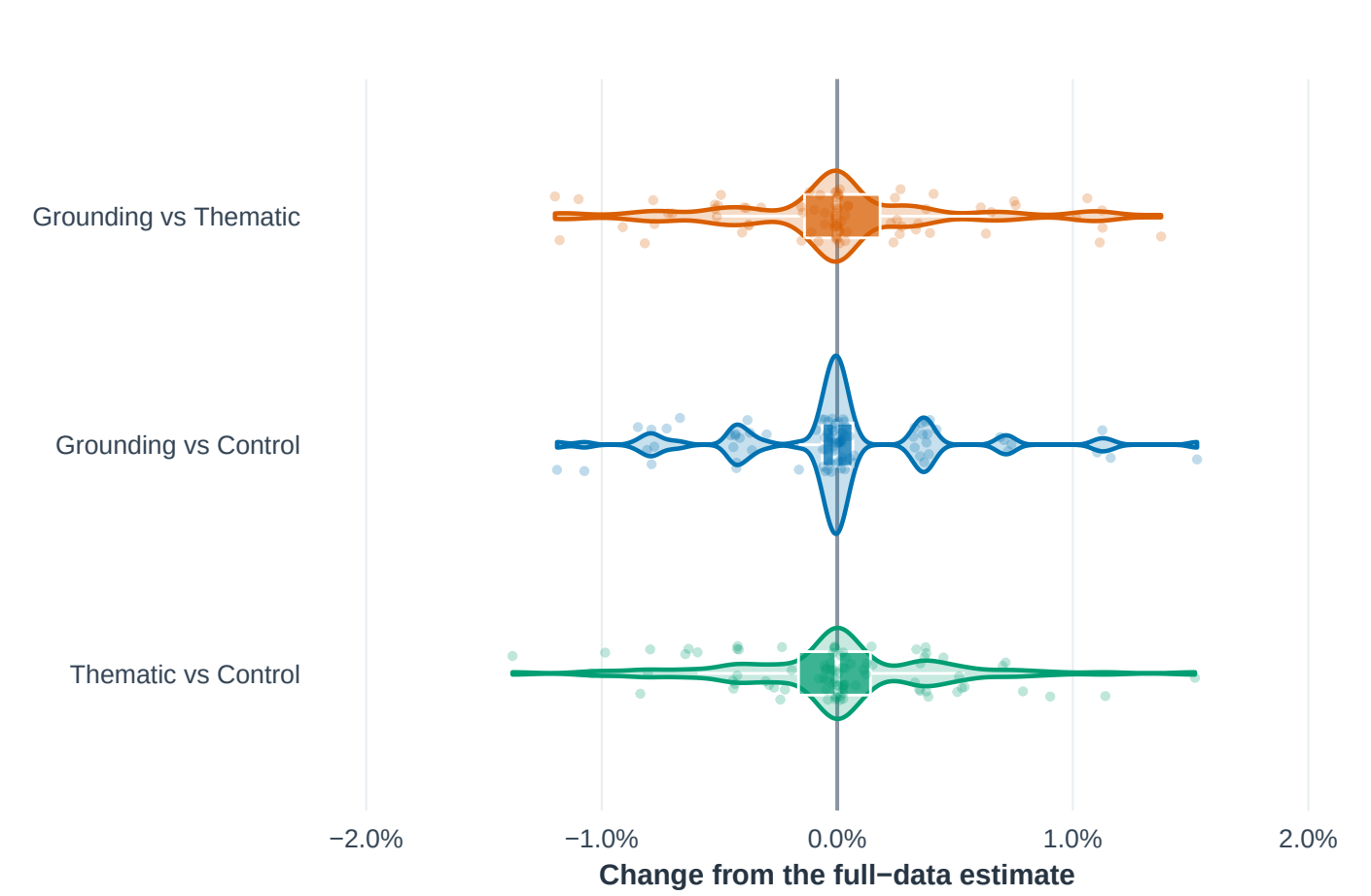
Each row refits the primary model after removing the named word



Dashed lines show full-data estimates. Thematic vs Control is item-sensitive.

E No single participant drives the result

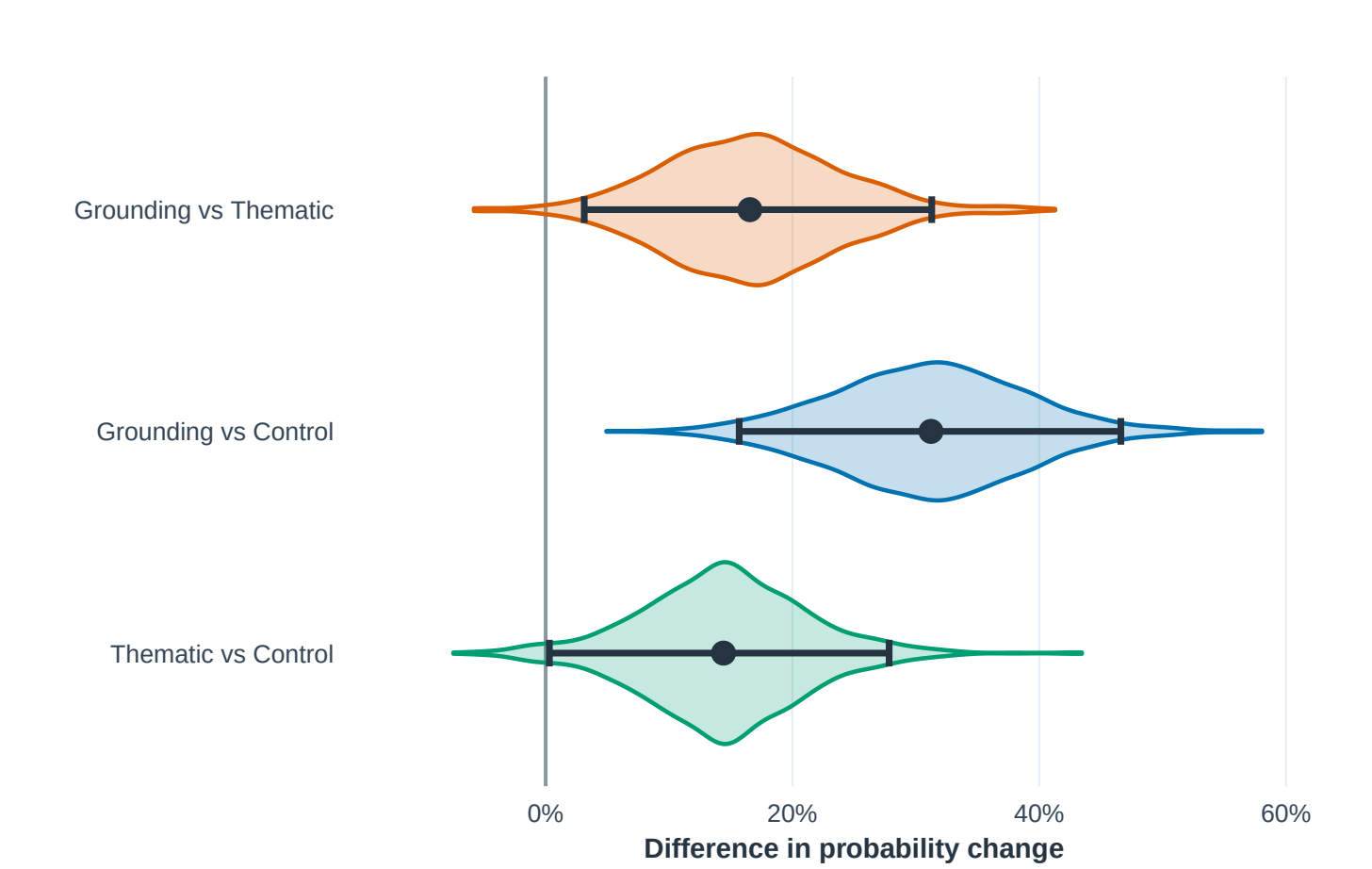
Movement in each contrast after deleting one participant and refitting the model



All 90 refits retained positive, Holm-significant contrasts.

F Crossed participant-item bootstrap

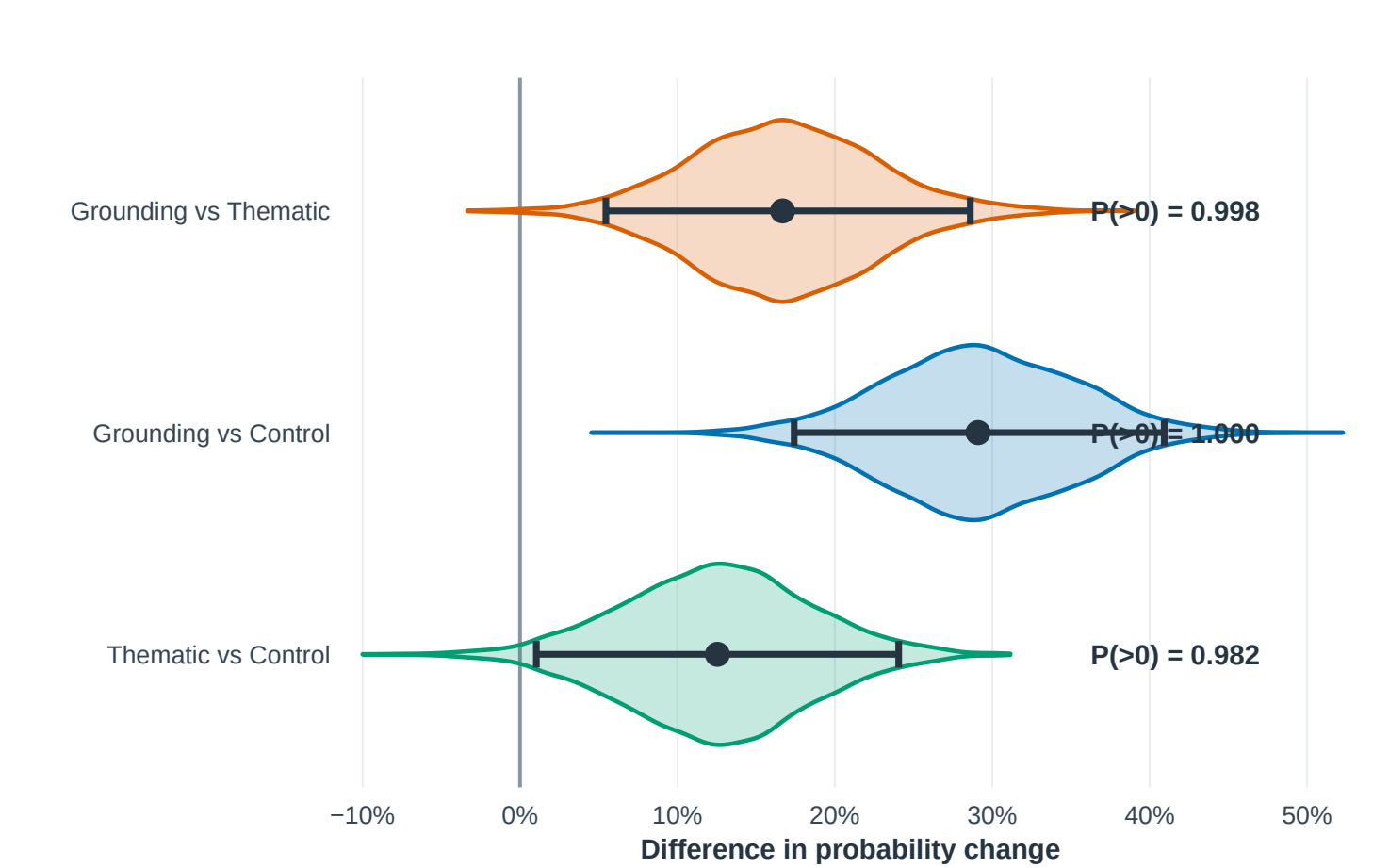
1,000 resamples of both participants and target words



Black intervals are percentile 95% intervals; points are bootstrap medians.

G Bayesian random-slopes sensitivity

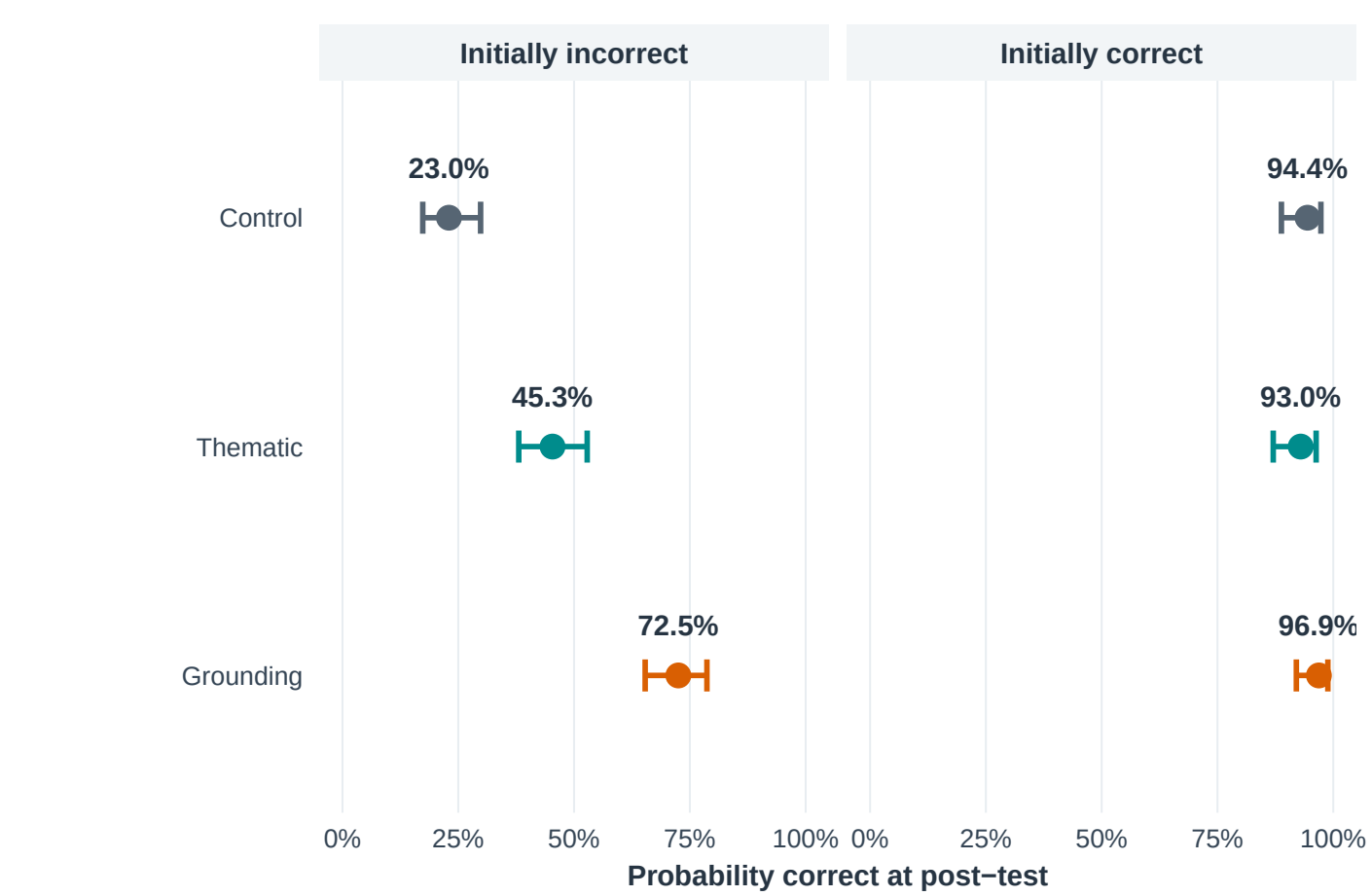
Posterior differences in pre-to-post probability change



Black intervals are 95% credible intervals. Four chains, 4,000 post-warmup draws, zero divergences.

H Post-test accuracy conditional on the pretest response

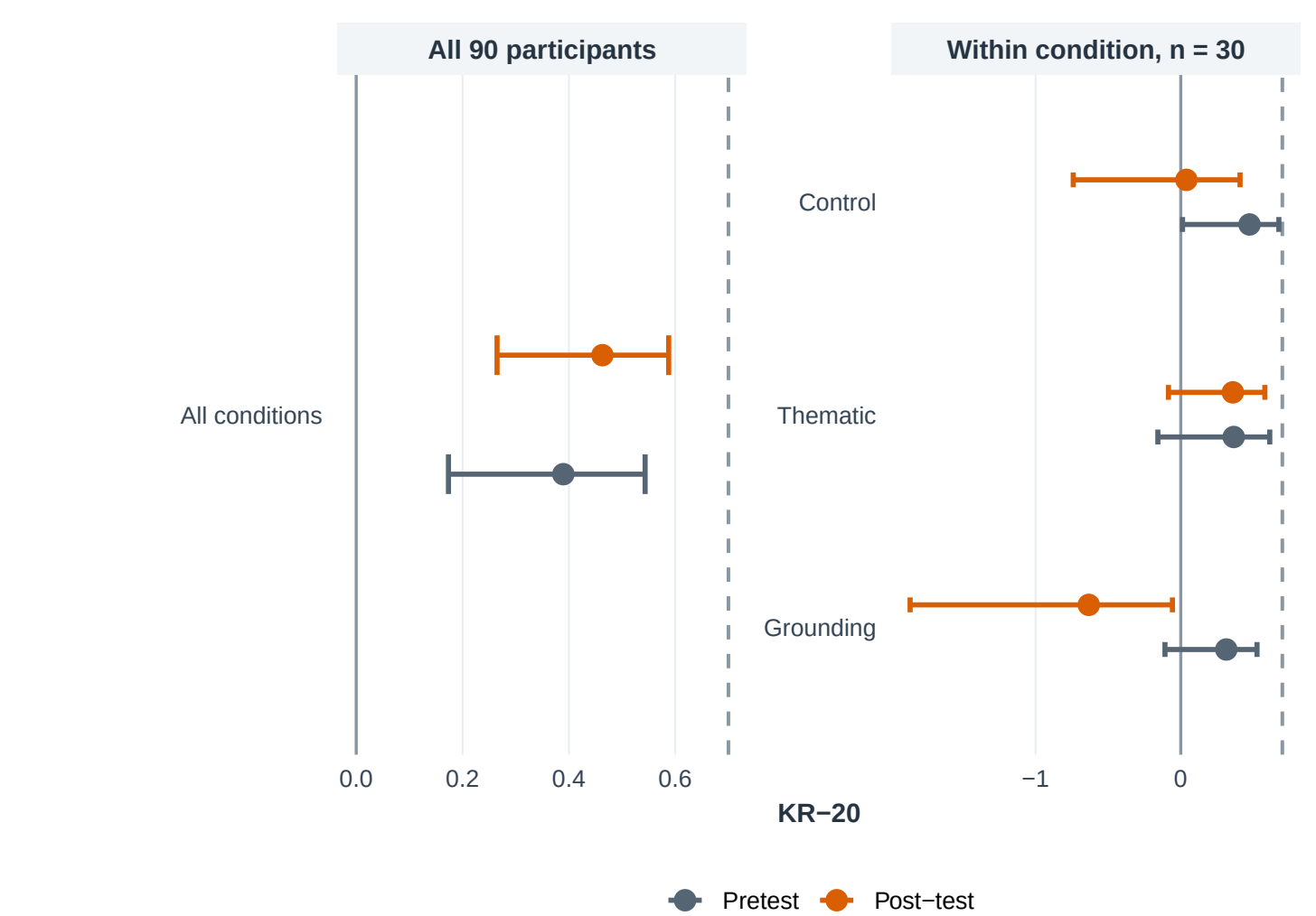
Conversion of initially incorrect responses differs sharply by condition



Response transitions are not pure knowledge-state transitions because question format changed. The exploratory model was singular.

I The 10-item score has weak internal consistency

KR-20 estimates with participant-bootstrap 95% intervals



The dashed line marks .70. The negative Grounding post-test estimate accompanies a strong ceiling.